

B.Tech I Year I Semester (R23) Regular Examinations January 2024

LINEAR ALGEBRA & CALCULUS

(Common to CS&IT, CSE (AI&ML), CSE (AI), CSE (CS), CSE (DS), CSE (IoT), AI&DS, AI&ML, CS&D, AI, CE, ECE, ME, IT, EE(VLSI&T), EEE and CSE)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

- (a) State Cauchy–Binet formulae. 2M
- (b) Define Diagonally dominance property. 2M
- (c) Find the Eigen values of the Matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$. 2M
- (d) Find the quadratic form relating to the symmetric matrix $\begin{bmatrix} 1 & 3 & -1 \\ 3 & 4 & 5 \\ -1 & 5 & 2 \end{bmatrix}$. 2M
- (e) Verify Cauchy's mean value theorem for $f(x) = x^2$, $g(x) = x^3$ in $[1, 2]$. 2M
- (f) The value of c of the Rolle's Theorem for the function $f(x) = \frac{x^3}{3} - 3x$ in the interval $[0, 3]$ is ----- 2M
- .
- (g) If $u = x^y$ then $\frac{\partial u}{\partial x} = \text{-----}$. 2M
- (h) If $x = r \cos \theta$, $y = r \sin \theta$ then $J\left(\frac{x, y}{u, v}\right) =$ 2M
- (i) $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} dx dy dz =$ 2M
- (j) Convert $\int_0^a \int_0^{\sqrt{a^2-x^2}} (x^2 + y^2) dy dx$ in polar coordinates. 2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Show that the three equations: 10M
- $$\begin{aligned} -2x + y + z &= a \\ x - 2y + z &= b \\ x + y - 2z &= c \end{aligned}$$
- have no solution unless $a + b + c = 0$ and in which case they have infinitely many solutions. Find these solutions when $a = 1$, $b = 1$, $c = -2$.
- OR**
- 3 Solve the following system of equations by Gauss Elimination method. 10M
- $$2x + y + 2z + w = 6; \quad 6x - 6y + 6z + 12w = 36; \quad 4x + 3y + 3z - 3w = -1; \quad 2x + 2y - z + w = 10.$$
- 4 Find a matrix P which diagonalizes A . verify that $P^{-1}AP = D$, and find A^4 10M
- $$\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}.$$
- OR**
- 5 Find the inverse of the Matrix $\begin{bmatrix} 1 & 3 & -1 \\ 3 & 4 & 5 \\ -1 & 5 & 2 \end{bmatrix}$ by using Cayley Hamilton theorem. 10M

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- 6 (a) Verify Lagrange's Mean value theorem for the function $f(x) = \cos x$ in the interval $[0, \pi/2]$. 5M
 (b) If $f(x) = \log x$, $g(x) = x^2$ in $[a, b]$ with $b > a > 1$, then using Cauchy's mean value theorem find the value of c . 5M

OR

- 7 (a) Expand $\log \sin x$ about $x = 3$ using Taylor's theorem. 5M
 (b) Obtain the Maclaurin's expansion of the function $\sinh x$. 5M
- 8 (a) If $U = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, $x^2 + y^2 + z^2 \neq 0$ then prove that $\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} = 0$. 5M
 (b) Find the minimum value of $x^2 + y^2 + z^2$ given that $xyz = a^2$. 5M

OR

- 9 (a) Show that the rectangular solid of maximum volume that can be inscribed in a sphere is a cube. 5M
 (b) Verify $u = 2x - y + 3z$, $v = 2x - y - z$, $w = 2x - y + z$ are functionally dependent and if so, find the relation between them. 5M

- 10 Change the order of integration and evaluate $\int_0^\infty \int_0^x x e^{-x^2/y} dy dx$. 10M

OR

- 11 Evaluate $\iiint xyz dx dy dz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$. 10M
